🔧 Nested Virtualization Lab Infrastructure: A to Z Guide

🎯 Objective

Set up a lab environment on a Windows 11 Host with:

- Hyper-V enabled
- A Windows Server 2022 VM as Domain Controller (DC)
- Internal NAT for internet access
- Multiple nested VMs (Windows Server 2019, Windows 11, Ubuntu, etc.)
- All client VMs joined to the domain and have internet access

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🖥️ Step 1: Host Machine Preparation

Requirements:

- Windows 11 Pro
- Hyper-V enabled
- Virtualization enabled in BIOS
- At least 32 GB RAM, 100+ GB SSD
- Internet access via Wi-Fi or Ethernet

✔️ Enable Hyper-V (if not already):

- Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All

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🌐 Step 2: Create Network Adapters in Hyper-V

Create Internal Network (for nested VMs):

1. Open **Hyper-V Manager > Virtual Switch Manager**
2. Create **Internal** switch → Name: Internal

Create External Network (for host Internet access):

1. Also create **External** switch bound to your **Wi-Fi/Ethernet**

🖥️ Step 3: Install and Configure Windows Server 2022 as Domain Controller

1. Create a Gen 2 VM: HOMEDC01
2. Attach to **Internal switch**
3. Allocate 4 GB RAM, 2 vCPU, 60 GB HDD
4. Install Windows Server 2022
5. Set a static IP:

   PowerShell:

- New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.200.1 -PrefixLength 24
- Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 127.0.0.1

Install AD DS & Promote to Domain Controller:

PowerShell

- Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
- Install-ADDSForest -DomainName "homedc.pvt"

🌐 Step 4: Setup NAT for Internet Access

Assign static IP and create NAT:

PowerShell

- New-NetIPAddress -IPAddress 192.168.200.1 -PrefixLength 24 -InterfaceAlias "Ethernet"
- New-NetNat -Name "InternalNAT" -InternalIPInterfaceAddressPrefix 192.168.200.0/24

🧱 Step 5: Create and Configure Client VMs

Repeat for:

- Windows Server 2019
- Windows 11
- Ubuntu or CentOS

Network:

- Attach to **Internal Switch**

- Assign static IP (example for Win Server 2019):

PowerShell

- New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.200.11 -PrefixLength 24 -DefaultGateway 192.168.200.1
- Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 192.168.200.1

🧪 Step 6: Join Clients to Domain

For Windows clients:

PowerShell

- Add-Computer -DomainName "homedc.pvt" -Credential homedc\Administrator -Restart

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🌍 Step 7: Enable IP Forwarding (optional for routing)

PowerShell

- Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters" -Name "IPEnableRouter" -Value 1

Restart-Computer

🧪 Step 8: Post-Setup Verification Script

Create a script to test everything after reboot:

PowerShell

Write-Host "=== Verifying IP Configuration ==="

Get-NetIPAddress | Where-Object {$_.AddressFamily -eq "IPv4"}

Write-Host "`n=== Testing Ping to Gateway (192.168.200.1) ==="

ping 192.168.200.1

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Write-Host "`n=== Testing Internet Access (Google DNS) ==="
ping 8.8.8.8

Write-Host "`n=== Testing DNS Resolution ==="
ping google.com
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